Data Analyst Project

Job Vacancy Tweets Analysis

Muhammad Farel Daffa
June 2026

Table of Contents

About Me

Profile

Muhammad Farel Daffa, S.Si.

Data Analyst

Experiences

Education

Bachelor of Science in Astronomy | Institut Teknologi Bandung
Aug 2021–Jul 2025

Experiences

Data Analyst Intern | Pikiran Rakyat Media Network
Nov 2025–Jun 2026

Research Intern | Badan Riset dan Inovasi Nasional
Mar–Apr 2025

Head of Member Resource | Himpunan Mahasiswa Astronomi ITB
Mar 2024–Feb 2025

Head of Research and Development | Himpunan Mahasiswa Astronomi ITB
Jun–Sep 2023

Project Overview

Understanding the business problem, objectives, and analytical framework

Business Understanding

Twitter has become a popular platform for companies to share job vacancies and reach potential candidates.

Analyzing companies, job positions, engagement, and posting patterns helps uncover hiring trends in the digital labor market.

Objectives

Identify the most active users based on posting volume and engagement during the observation period.

Analyze job posting behavior across year, day of week, and hour to uncover productivity patterns and engagement timing.

Identify the most frequently advertised job positions and their engagement levels.

Framework

Data Collection

Obtain the Job Vacancy Tweets dataset from Kaggle and explored 50,000 rows × 13 columns.

Data Preparation

Process, transform, clean, and standardize the raw dataset into an analysis-ready format.

Exploratory Data Analysis

Identify hiring trends, posting behavior, and engagement patterns in job vacancy tweets.

Dashboard Development

Build an interactive dashboard in Google Data Studio.

Insights Generation

kaggle

Dataset Overview

Overview dataset structure, quality, and variables

Dataset Size	Period
50,000 × 13	2019–2023

Total Missing Values	Total Duplicate Rows
15,305	0

Source	Author
Kaggle	Prasad Patil

Additional Notes

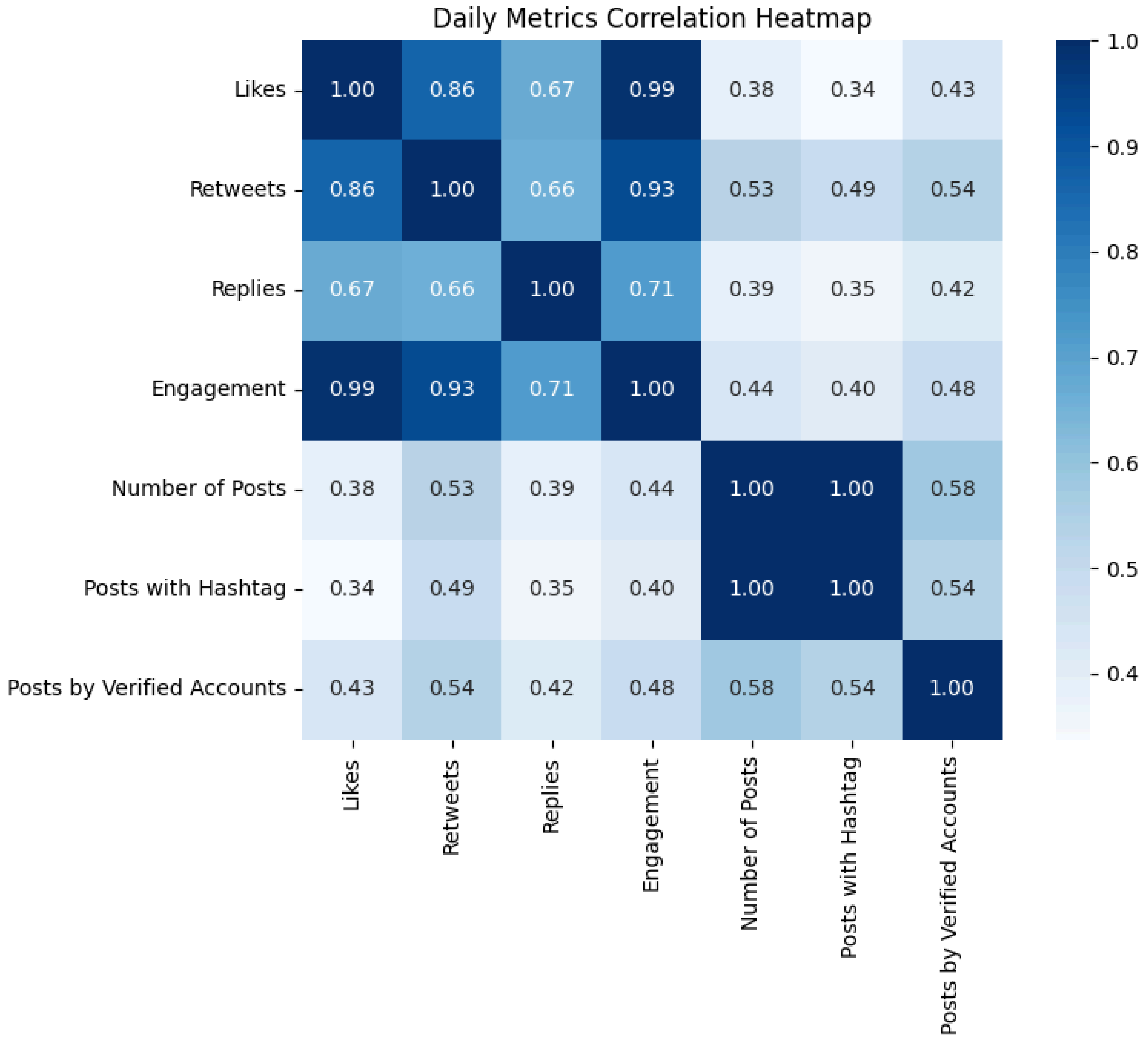
- Engagement is sum of Retweets, Likes, and Replies.
- Location isn’t used in this analysis because it’s highly inconsistent and cannot reliably represent users' actual geographic locations.
- NaN data from hashtag column will not be removed.

Variables	Description
ID	The unique identifier of the tweet
Timestamp	The date and time when the tweet was posted
User	The Twitter handle of the user who posted the tweet
Text	The content of the tweet
Hashtag	The hashtags included in the tweet, if any
Retweets	The number of retweets as of the time it was scraped
Likes	The number of likes as of the time it was scraped
Replies	The number of replies as of the time it was scraped
Source	The source application or device used to post the tweet
Location	The location listed on the user's Twitter profile, if any
Verified_Account	A Boolean value whether the user's account has been verified
Followers	The number of followers as of the time the tweet was scraped
Following	The number of following as of the time the tweet was scraped

Exploratory Data Analysis

Determine correlation between daily metrics

Heatmap Correlation



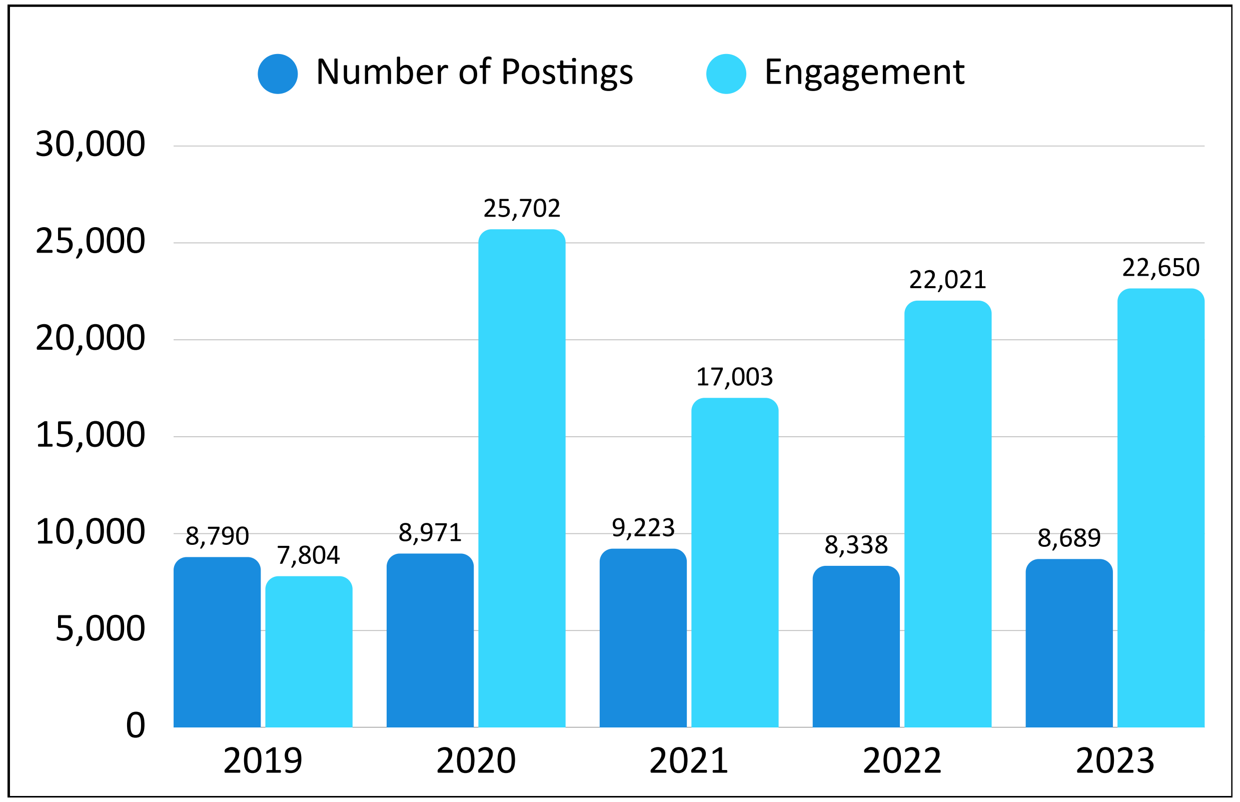
Insights

- Posting volume has a moderate correlation with engagement (r = 0.44), suggesting that **content quality and relevance matter more than posting frequency**.
- Verified accounts show a stronger correlation with engagement (r = 0.48) than posting volume, highlighting the **importance of account credibility**.
- Hashtag usage has a relatively weak correlation with engagement (r = 0.40), indicating **it supports discoverability but isn't a key engagement driver**.

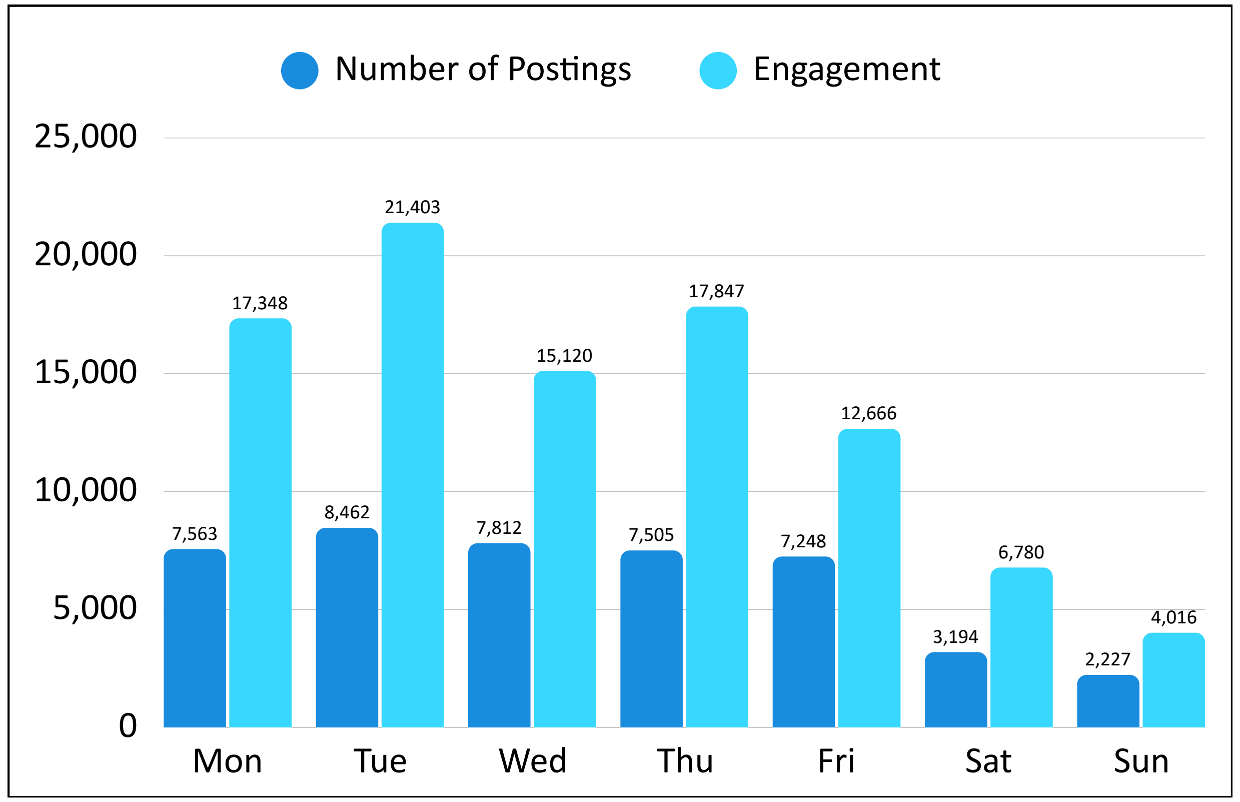
Exploratory Data Analysis

Identify posting behavior in job vacancy tweets by quantity and engagement

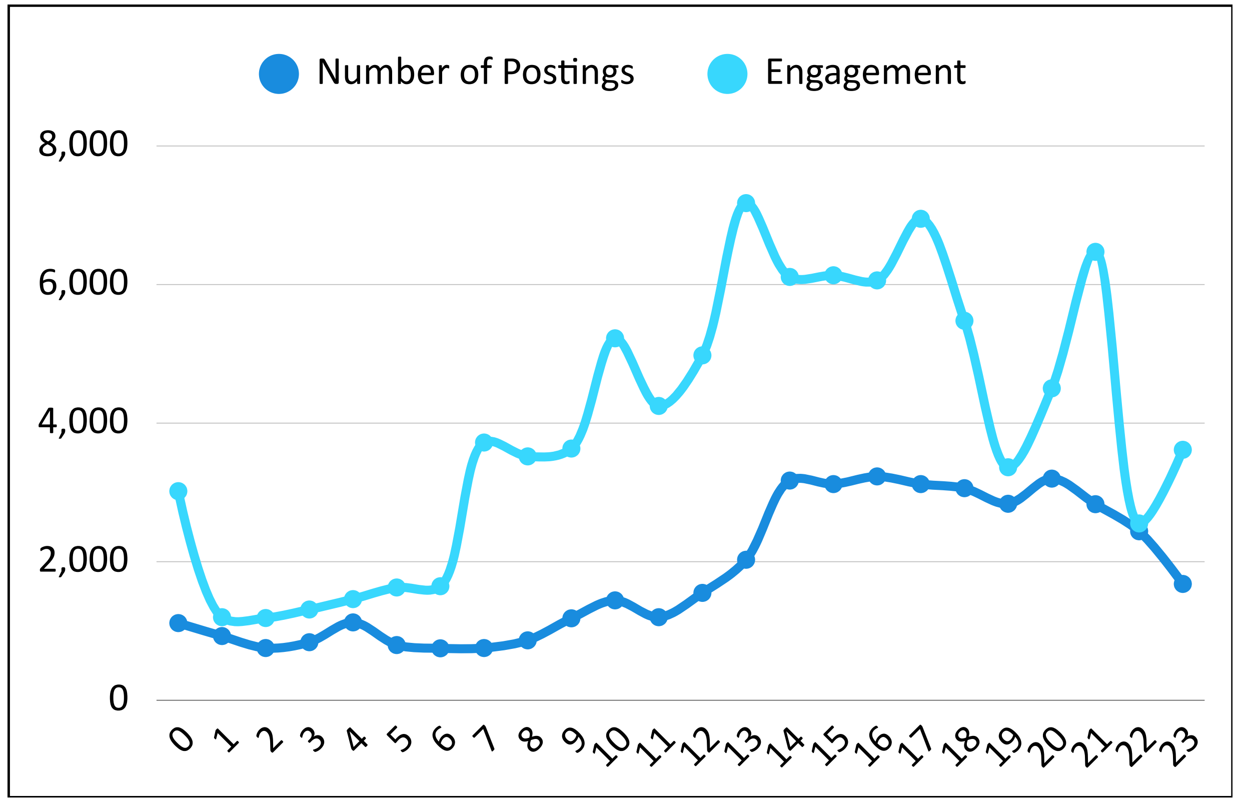
Number of Postings per Year



Number of Postings per Day of the Week



Number of Postings per Hour



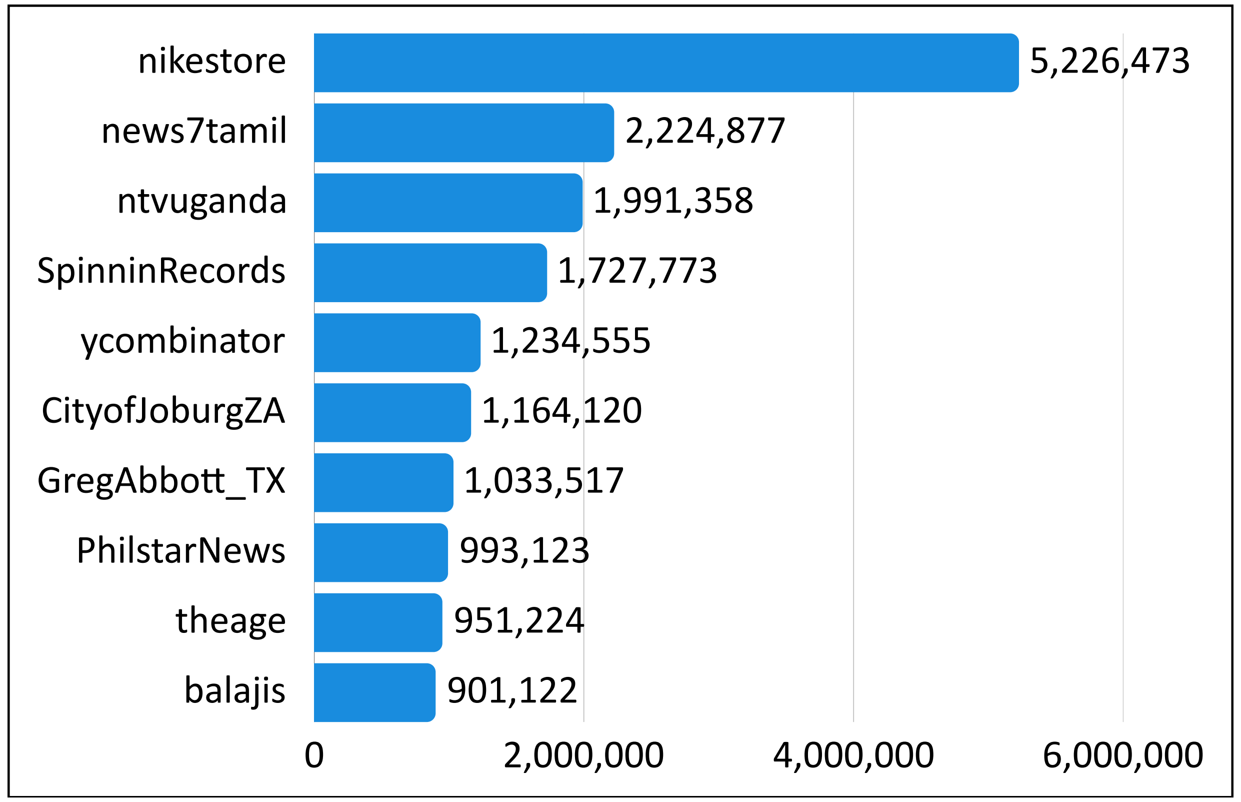
Insights

- Posting activity remained relatively stable (±8.8 tweets/year). Engagement fluctuated significantly, indicating **engagement is influenced more by content performance than posting volume**.
- **Weekday posts consistently outperformed weekends**, with Tuesday generating the highest engagement (±21.4K), while weekeends recorded the lowest posting activity and engagement.
- **Posting volume was concentrated between 14:00–21:00**, suggesting this is the primary publishing window for discussions and content distribution.
- **Engagement efficiency concentrated between 13:00–17:00 and at 21:00**, indicating stronger audience responsiveness during these periods.

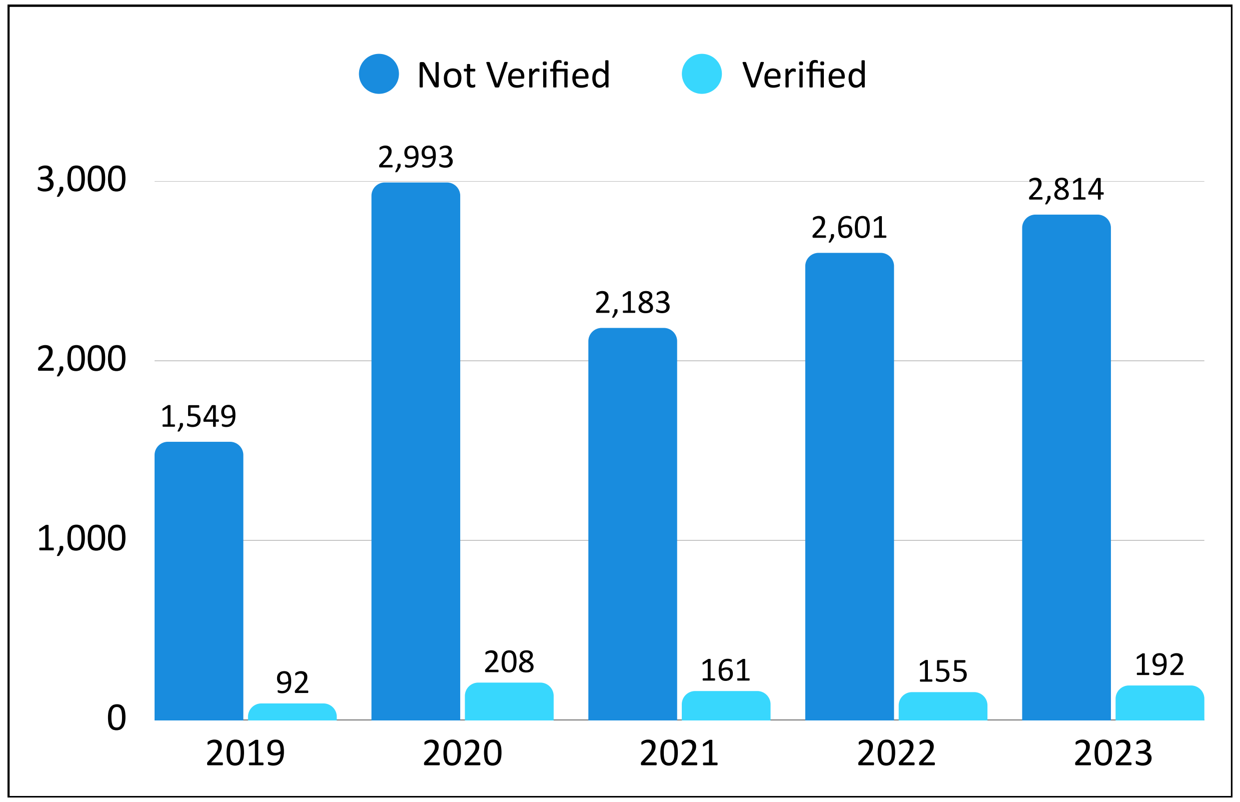
Exploratory Data Analysis

Identify user behavior in job vacancy tweets

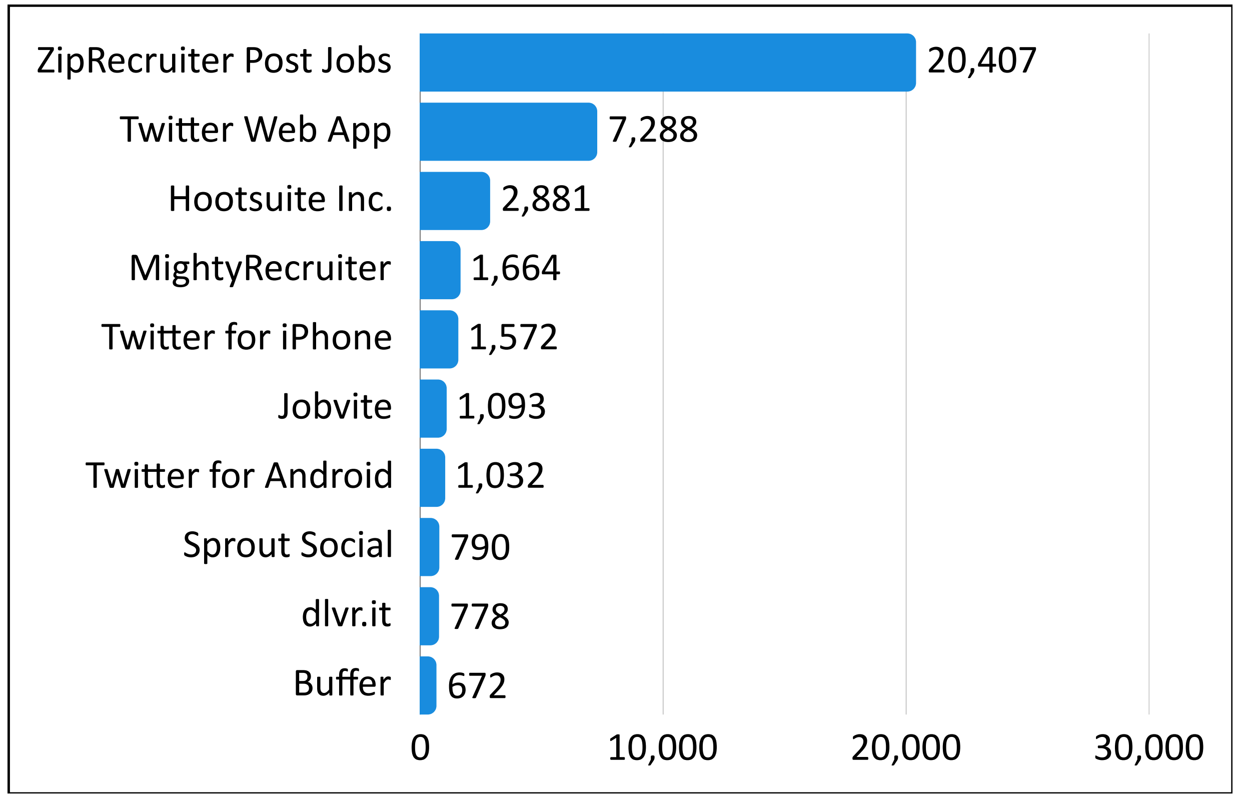
Top 10 User by Followers



Number of Unique Users by Account Status



Top 10 Sources



Insights

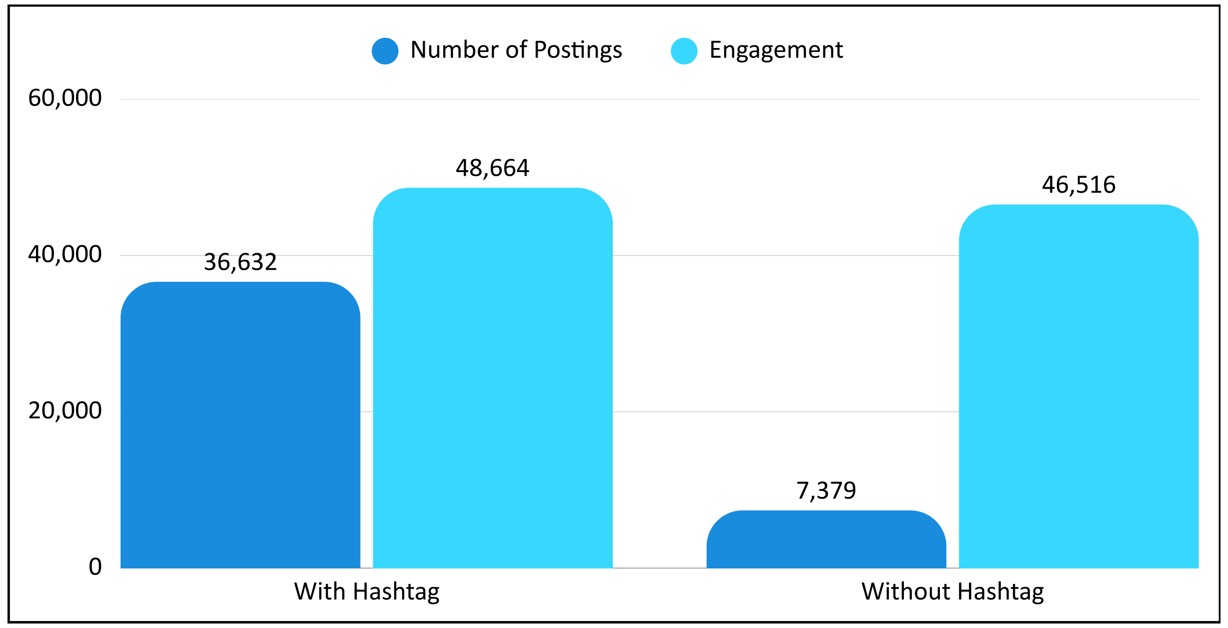
- **Conversation volume was dominated by non-verified accounts across all years (±93.8%)**, indicating discussions were primarily driven by the broader user community.
- **Job vacancies were primarily posted by high-reach accounts**, with several recruiters, media outlets, and organizations having follower bases exceeding 900K, helping expand visibility.
- **Posting activity was heavily driven by third-party publishing tools**, indicating a strong reliance on automated content distribution rather than purely organic posting.

Exploratory Data Analysis

Identify content in job vacancy tweets

Top 50 WordCloud

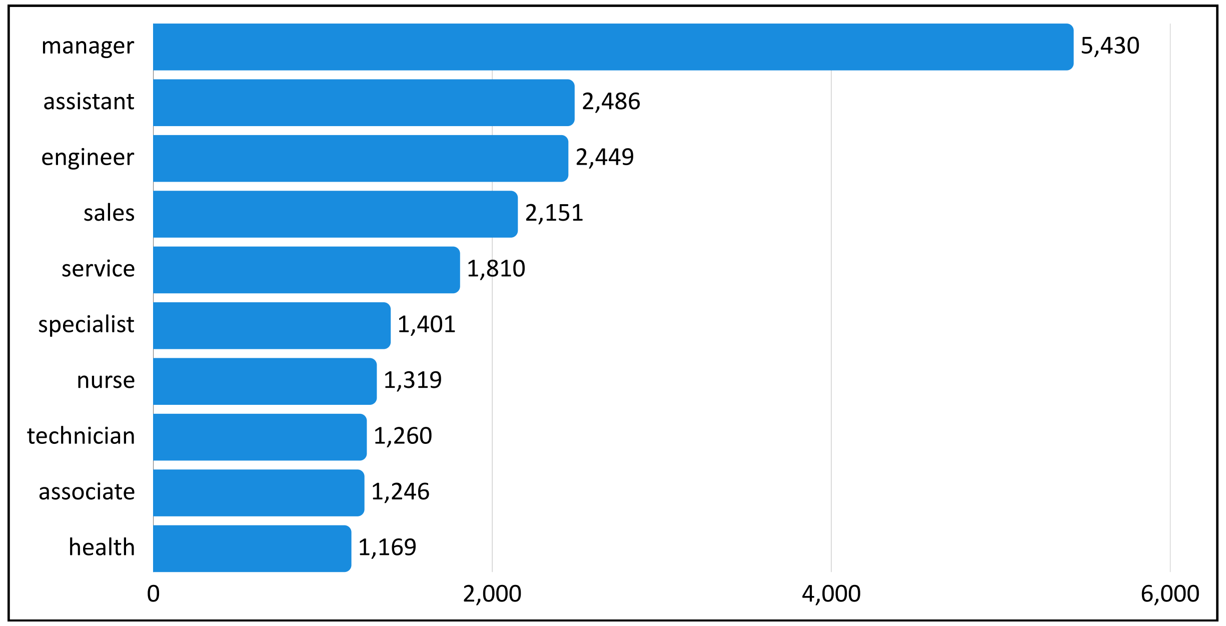
Hashtag Analysis



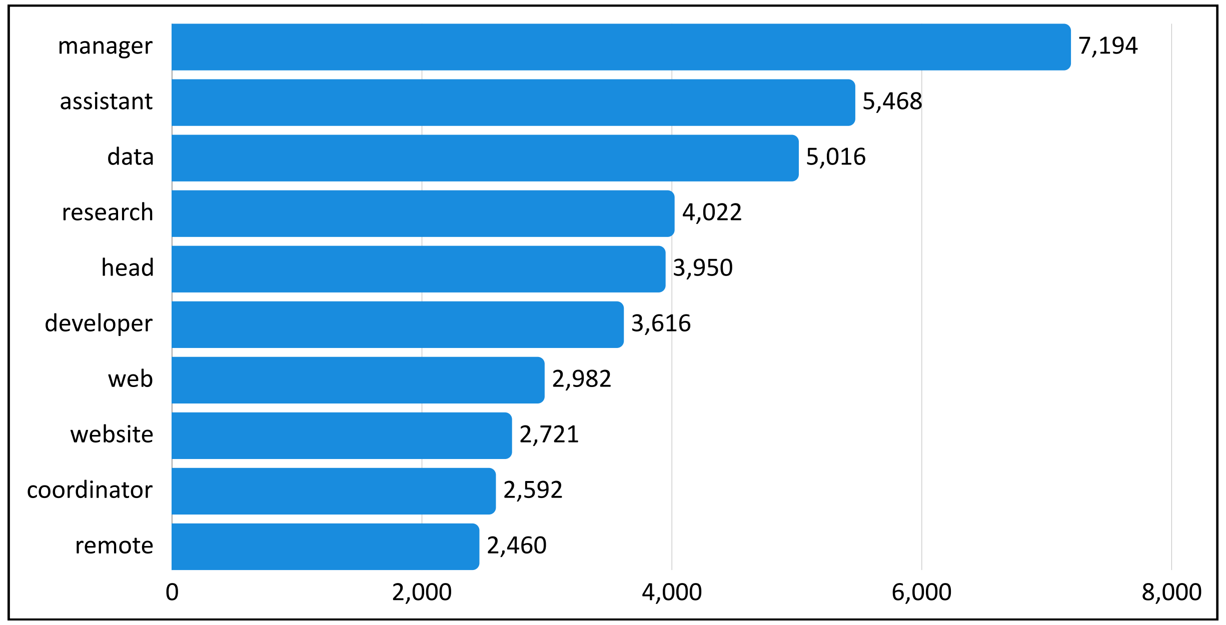
Insights

- Tweets with hashtags slightly outperformed tweets without them in engagement, indicating **hashtags are one of a key engagement driver**.
- Manager and assistant were the most frequently mentioned and engaged roles in job vacancy tweets, showing **strong demand for this positions**.

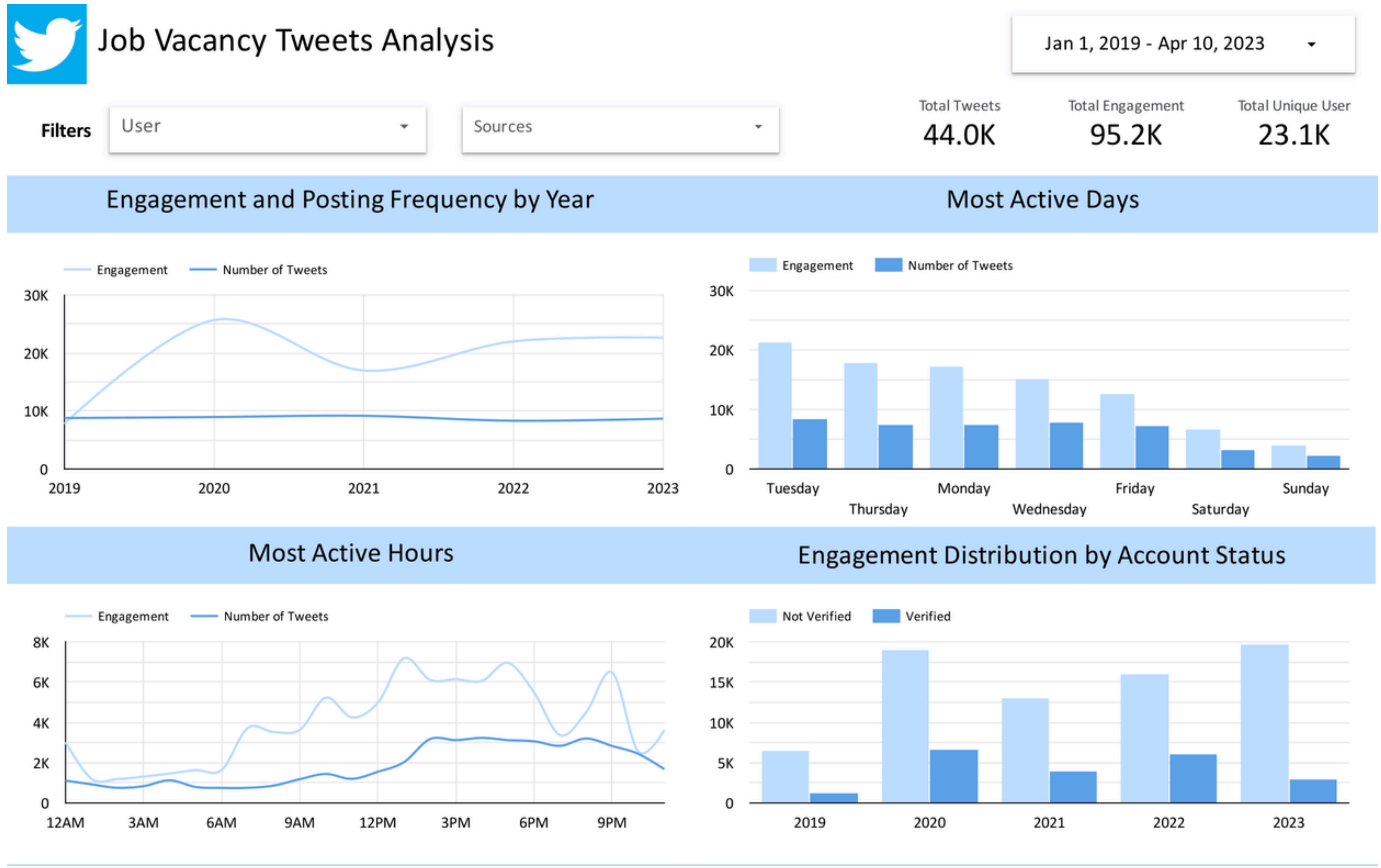
Top 10 The Most Looking Job and Field

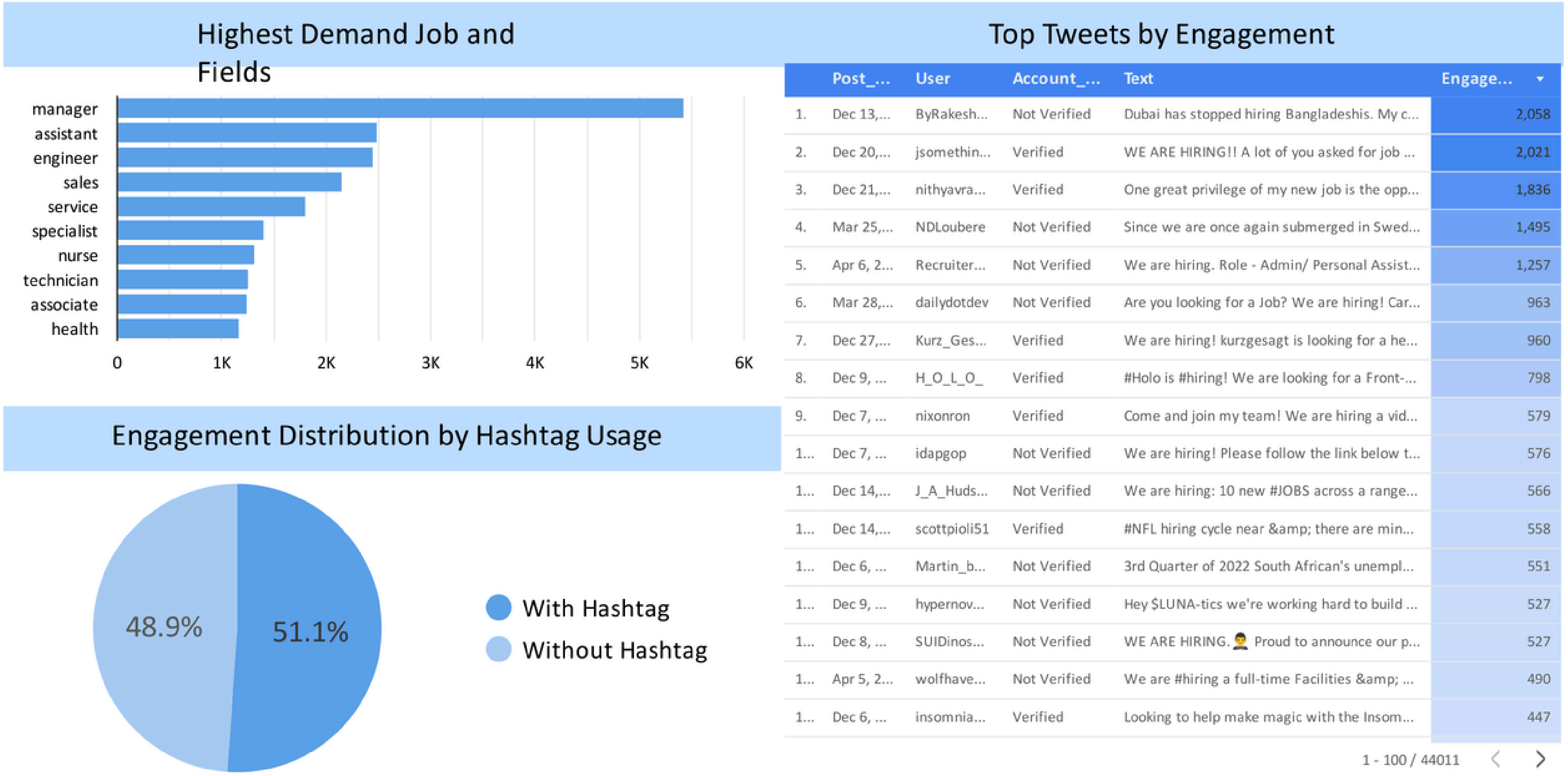


Top 10 Job and Field by Engagement



Dashboard Preview





	Post_...	User	Account_...	Text	Engage...
1.	Dec 13,...	ByRakesh...	Not Verified	Dubai has stopped hiring Bangladeshis. My c...	2,058
2.	Dec 20,...	jsomethin...	Verified	WE ARE HIRING!! A lot of you asked for job ...	2,021
3.	Dec 21,...	nithyavra...	Verified	One great privilege of my new job is the opp...	1,836
4.	Mar 25,...	NDLoubere	Not Verified	Since we are once again submerged in Swed...	1,495
5.	Apr 6, 2...	Recruiter...	Not Verified	We are hiring. Role - Admin/ Personal Assist...	1,257
6.	Mar 28,...	dailydotdev	Not Verified	Are you looking for a Job? We are hiring! Car...	963
7.	Dec 27,...	Kurz_Ges...	Verified	We are hiring! kurzgesagt is looking for a he...	960
8.	Dec 9, ...	H_O_L_O_	Verified	#Holo is #hiring! We are looking for a Front-...	798
9.	Dec 7, ...	nixonron	Verified	Come and join my team! We are hiring a vid...	579
1...	Dec 7, ...	idapgop	Not Verified	We are hiring! Please follow the link below t...	576
1...	Dec 14,...	J_A_Huds...	Not Verified	We are hiring: 10 new #JOBS across a range...	566
1...	Dec 14,...	scottpioli51	Verified	#NFL hiring cycle near & there are min...	558
1...	Dec 6, ...	Martin_b...	Not Verified	3rd Quarter of 2022 South African's unempl...	551
1...	Dec 9, ...	hypernov...	Not Verified	Hey $LUNA-tics we're working hard to build ...	527
1...	Dec 8, ...	SUIDinos...	Not Verified	WE ARE HIRING. 🧑‍💼 Proud to announce our p...	527
1...	Apr 5, 2...	wolfhave...	Not Verified	We are #hiring a full-time Facilities & ...	490
1...	Dec 6, ...	insomnia...	Verified	Looking to help make magic with the Insom...	447

[Link Dashboard]

Recommendations

Transform data into business recommendations and actionable insights

Recommendations	Actionable Insights	Expected Impact
Publish During Prime Day and Time	Prioritize posting on Tuesday–Thursday and active working time, especially at 13:00–17:00.	Increase tweet visibility and median engagement by reaching the audience when they are most active.
Increase the Use of Hashtag	Increase the use of 1–3 hashtag in tweets, such as #Hiring, #NameoftheJob, #NameoftheCompany.	Improve content discoverability and organic reach, attracting a broader audience and increasing engagement.
Focus on Quality Content	Focus on content quality and audience relevance rather than simply increasing posting volume.	Improve engagement efficiency by generating higher interactions per tweet.
Focus on High Demand Job	Develop more content around high-engagement job categories, such as manager and data, research, and developer roles.	Increase engagement by publishing content that better matches audience demand, resulting in more interactions and potential applicants.

LET'S CONNECT

https://www.linkedin.com/in/muhammad-farel-daffa

fareldf4@gmail.com

+6281211999109